GAVIN FAUS

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EDUCATION

University of Denver – Denver, CO Expected Graduation: November 2027

Bachelor of Science in Mechanical Engineering | Minors: Mathematics, Physics | GPA: 3.06

Relevant Coursework: CAD Design & Analysis, Materials Science, Engineering Design, Mechatronics, Fluid Dynamics, Heat Transfer, Thermodynamics

ENGINEERING PROJECT EXPERIENCE

Senior Design Project – Charger Packaging & Thermal Management Lead Sep 2026 – Present

- Lead mechanical packaging and thermal management for a five-member engineering team, developing enclosure mounting, airflow paths, ducting, and fan/blower cooling solutions.
- Define thermal requirements based on component heat losses and enclosure conditions; develop temperature verification plans, sensor placement, and controlled cooling-fault testing procedures.

Transient Thermal Modeling & Material Identification March – June 2026

- Developed **three transient heat-transfer models** in Excel to simulate cooling of an **unknown metallic block**, comparing aluminum, steel, and tin candidate materials across convection and radiation models; calibrated emissivity (**ε ≈ 0.127**) via least-squares fitting, achieving **1.62% final-temperature error for aluminum**.
- Built a **3D transient thermal FEA model of the block** in SOLIDWORKS, running aluminum, steel, and tin material configurations with consistent geometry and meshing; compared transient temperature responses against experimental data and identified **aluminum** as the closest match.

Autonomous Transport & Placement System March – June 2026

- Designed, built, and tested a custom autonomous vehicle integrating mechanical hardware, sensors, actuators, and controls for payload classification, path following, obstacle avoidance, and transport across a **3.8 m course**.
- Developed closed-loop C++ control logic on a Teensy 4.1 and integrated color/ultrasonic sensors, DC motors, servos, and 3D-printed hardware to support autonomous navigation and obstacle avoidance.

Thermoelectric Candle-Powered Generator March – June 2025

- Designed, fabricated, and tested a **thermoelectric energy-conversion system** using a Peltier module to convert candle heat into electrical energy and power a DC motor, producing **~0.8 V**.
- Built, tested, and iteratively improved **cold-side cooling, motor selection, gearing, and rolling resistance;** verified that the final prototype transported a payload **11 ft**, **exceeding the 6 ft travel requirement**.

Converging Nozzle CFD Analysis Jan – March 2026

- Created a **converging nozzle** in SOLIDWORKS and modeled steady, incompressible oxygen flow using **SOLIDWORKS Flow Simulation**, applying a **2.5 m/s inlet velocity**, **1 atm outlet pressure**, and **no-slip wall boundary conditions**.
- Analyzed velocity trajectories and static-pressure distributions, observing flow acceleration and pressure reduction through the converging section; compared CFD-predicted performance trends with continuity and Bernoulli principles to confirm physically consistent behavior.

ENGINEERING ACTIVITIES

University of Denver Time Attack Racing – Member Jan 2026 – Present

- Investigated CFD methodologies in **SOLIDWORKS Flow Simulation** to support future cooling-system development for high-altitude racing conditions.
- Initiated research into supplemental driver oxygen systems and authored a technical brief evaluating high-altitude hypoxia mitigation strategies.

TECHNICAL SKILLS

Mechanical Design & Simulation: SOLIDWORKS, SOLIDWORKS Simulation (FEA), SOLIDWORKS Flow Simulation (CFD), ANSYS Mechanical, ANSYS Fluent

Prototyping & Testing: 3D Printing, Tensile Testing, Charpy Impact Testing, Rockwell Hardness Testing, Metallographic Preparation, Optical Microscopy, Infrared Thermography

Programming & Data Analysis: MATLAB, Excel, Python, C++, Arduino, Teensy 4.1, Sensors & DAQ